Final Project: API-Integrated System

Overview

Each student will individually design, build, and present a working system that integrates **at least 5 external APIs** into one purposeful solution. Final submission of the project will be on **October 26 to 30, 2026**. The project should solve a real problem for a clear set of users. You are not required to pay for an API.

Core system features

- User registration and login, with at least two user roles (e.g., admin and regular user)
- A database that stores the system's own records (MySQL, PostgreSQL, Firebase, Supabase, or similar)
- At least one full CRUD module
- A dashboard or summary page for the admin role

Example API combinations

System idea	Possible APIs
Travel planner	Google Maps / OpenStreetMap, OpenWeatherMap, ExchangeRate API, Amadeus, SendGrid
Online clinic booking	Google Calendar, PayMongo sandbox, Semaphore SMS, Google OAuth, Cloudinary
Local market e-commerce	PayPal sandbox, Mapbox, Mailgun, Cloudinary, OpenAI / Gemini
Disaster readiness portal	OpenWeatherMap, USGS Earthquake, Leaflet + OSM, Twilio, NewsAPI

These are examples only. Original ideas are encouraged.

Scope and technology

- **Individual work:** this is an individual project. Each student builds, documents, and presents their own system. Group work and shared code between students are not allowed.
- **Platform:** web application, mobile application, or both.
- **Languages and frameworks:** your choice (e.g., PHP/Laravel, Node.js/Express, Python/Django or Flask, React).
- **Version control:** a GitHub or GitLab repository is required, with regular commits from the start of the project. The commit history is part of grading.
- **Hosting:** deploying online (Vercel, Render, Railway, Netlify, or similar). You are not required to pay for your hosting. Just use the free hosting.

Grading rubric

The project is graded out of 100 points. Your commit history and Q&A answers are used to confirm the work is your own.

Criteria	Excellent	Very Good	Good	Fair	Poor	Max Points
System Usability and Functionality	System is fully functional, user-friendly, error-free, and all features operate smoothly throughout the demonstration. **(20-17)**	System functions properly with only minor issues that do not affect usability. **(16-13)**	Most features work correctly, but some functions have minor issues or limitations. **(12-9)**	Several features exhibit errors or usability problems that affect user experience. **(8-3)**	System is incomplete, unstable, or many features do not function properly. **(4-1)**	**20**
Content (System Features)	All required features are implemented and exceed expectations in supporting project objectives. **(20-17)**	Most required features are implemented with only minor missing components. **(16-13)**	Core features are present but some important requirements are lacking. **(12-9)**	Several required features are incomplete or insufficiently implemented. **(8-3)**	Major required features are missing or non-functional. **(4-1)**	**20**

Criteria	Excellent	Very Good	Good	Fair	Poor	Max Points
Innovation and Technical Complexity	Demonstrates exceptional innovation, creativity, and advanced technical implementation beyond standard requirements. **(15-13)**	Shows strong innovation and effective use of appropriate technologies. **(12-10)**	Demonstrates adequate innovation and acceptable technical implementation. **(9-7)**	Limited innovation with basic technical implementation. **(6-4)**	Minimal innovation and weak technical sophistication. **(3-1)**	**15**
Organization of Presentation	Presentation is highly organized, engaging, professional, and follows a logical sequence. **(10)**	Presentation is organized with minor lapses in flow or transitions. **(8)**	Presentation is understandable but lacks consistency in delivery and organization. **(6)**	Presentation is difficult to follow due to poor organization. **(4)**	Presentation lacks structure, preparation, and clarity. **(2)**	**10**
System Reliability and Performance	System demonstrates excellent performance, stability, speed, and reliability under all tested scenarios. **(10)**	System performs reliably with only minor performance issues. **(8)**	System generally performs well but exhibits occasional delays or minor issues. **(6)**	Performance issues are noticeable and affect demonstration quality. **(4)**	Frequent crashes, errors, or performance problems occur. **(2)**	**10**
Alignment with Objectives and Problem Solving	Project completely addresses the identified problem and achieves all stated objectives. **(10)**	Project addresses most objectives and effectively solves the identified problem. **(8)**	Project addresses the main objectives but with some limitations. **(6)**	Project partially addresses objectives and problem requirements. **(4)**	Project fails to address major objectives or solve the identified problem. **(2)**	**10**
Understanding of the System and Concepts	Demonstrates comprehensive mastery of the system,	Demonstrates strong understanding with	Demonstrates adequate understanding of most	Demonstrates limited understanding of important concepts	Unable to explain key concepts,	**10**

Criteria	Excellent	Very Good	Good	Fair	Poor	Max Points
	technologies, methodologies, and design decisions. **(10)**	only minor gaps in explanation. **(8)**	concepts and components. **(6)**	and system components. **(4)**	methodologies, or system functions. **(2)**	
Ability to Respond to Questions	Answers are accurate, confident, analytical, and supported by evidence. **(10)**	Answers are mostly accurate and demonstrate good understanding. **(8)**	Answers basic questions adequately but lack depth. **(6)**	Answers are incomplete or require significant prompting. **(2)**	Unable to answer most questions satisfactorily. **(2)**	**10**
TOTAL						**100**